

Coding Set 4

Problem 1: Function Overloading - Area Calculator

Create a class Area with overloaded functions calculate() to find:

- Area of a square
- Area of a rectangle
- Area of a circle

Sample

```
calculate(5);    // Square
```

```
calculate(4,6);  // Rectangle
```

```
calculate(3.5);  // Circle
```

Problem 2: Function Overloading - Maximum Finder

Create a class Maximum with overloaded functions max() that find:

- Maximum of two integers
- Maximum of three integers
- Maximum of two floating-point numbers

Output Example

Max of 10 and 20 = 20

Max of 5, 8 and 3 = 8

Max of 3.2 and 4.5 = 4.5

Problem 3: Operator Overloading - Complex Number Addition

Create a class Complex having:

- Real part
- Imaginary part

Overload the + operator to add two complex numbers.

Example

$(3 + 4i) + (2 + 5i)$

Output: $5 + 9i$

Problem 4: Operator Overloading - Distance Addition

Create a class Distance with:

- Feet
- Inches

Overload the + operator to add two distances.

If inches exceed 12, convert them appropriately into feet.

Example

5 ft 8 in + 3 ft 9 in

Output:

9 ft 5 in

Problem 5: Static Members - Student Counter

Create a class Student.

Requirements:

- Each time an object is created, increment a static variable count.
- Display the total number of student objects created.

Sample Output

Student Created

Student Created

Student Created

Total Students = 3

Problem 6: Static Members - Bank Account Statistics

Create a class BankAccount.

Requirements:

- Account number
- Customer name
- Static variable totalAccounts

Display the total number of bank accounts created.

Challenge

Also display the total accounts using a static member function.

Problem 7: Friend Function - Largest Number

Create a class with two private integers.

Write a friend function that finds and displays the larger number.

Example

Numbers: 25 and 40

Largest = 40

Problem 8: Friend Function - Sum of Two Classes

Create two classes:

class A

class B

Each class contains one private integer.

Write a friend function that accesses both classes and calculates their sum.

Example

A = 10

B = 20

Sum = 30

Problem 9: Inline Function - Simple Interest

Create a class Interest.

Write an inline function:

calculateSI()

to compute:

$$SI = \frac{P \times R \times T}{100}$$

where:

- P = Principal
- R = Rate
- T = Time

Example

P = 10000

R = 5

T = 2

SI = 1000

Problem 10: Mixed Problem (Comprehensive)

Create a class Book.

Requirements:

1. Store:
 - Book ID
 - Book Name
 - Price
2. Use a **static member** to count total books.
3. Use an **inline function** to display discount price (10% discount).
4. Overload the > operator to compare prices of two books.
5. Use a **friend function** to display details of the costlier book.

Sample Output

Book 1 Price = 500

Book 2 Price = 700

Costlier Book:

ID: 102

Name: C++ Programming

Price: 700

Total Books = 2

Discounted Price = 630